

Birla Institute of Technology & Science, Pilani

First Semester 2024-25

Mid-Semester Examination (Saturday, 05 Oct.)

Course No. : EEE/ECE F311

Course Title: Communication Systems

Max. Marks : 50

Time : 90 Min.

Name:

ID No.

Instructions: Use notation consistently. You may use standard results or formulae by stating them explicitly. Provide key intermediate steps. Highlight simplified final answers in rectangular boxes. Please avoid overwriting, as overwritten answers will not be rechecked.

Q. 1. (i) Write down the general expression for a single-tone AM signal $s(t)$ in time domain. Express $s(t)$ as the sum of the carrier term, LSB term and USB term. Assume that the single-tone AM uses the following signals and parameters: message signal $A_m \cos(2\pi f_m t)$, unmodulated carrier signal $A_c \cos(2\pi f_c t)$, and modulation index μ . [2]

Ans. General expression for single-tone AM signal (in time domain):

$$s(t) = A_c (1 + \mu \cos(2\pi f_m t)) \cos(2\pi f_c t).$$

AM signal expression as the sum of three terms:

$$s(t) = A_c \cos(2\pi f_c t) + \frac{\mu A_c}{2} \cos(2\pi(f_c - f_m)t) + \frac{\mu A_c}{2} \cos(2\pi(f_c + f_m)t).$$

(ii) Consider a single tone AM signal $s(t) = \alpha \cos(400 \pi t) + \beta \cos(360 \pi t) + \beta \cos(440 \pi t)$ V, where α and β are real positive constants. Comparing with the three-term general expression for the AM signal obtained in (i): determine carrier frequency, lower side-band (LSB) frequency, upper side-band (USB) frequency. What is the bandwidth of the AM signal $s(t)$? What is the message signal frequency? [2.5]

Ans. Carrier frequency $f_c = 200$ Hz, LSB frequency $= (f_c - f_m) = 180$ Hz, USB frequency $= (f_c + f_m) = 220$ Hz.

AM signal bandwidth $2f_m = 40$ Hz. $\therefore$ Message signal frequency $f_m = 20$ Hz.

(iii) Refer to part (ii). Suppose the carrier power is 128 W and the power efficiency is 20 %. Determine the values of α , β , and μ . Express $s(t)$. What is the expression for μ as a function of α and β ? [5]

Ans. Carrier power $= \frac{A_c^2}{2} = 128$ W $\Rightarrow A_c = 16$ V $\Rightarrow \alpha = 16$.

Power efficiency $= 20$ % $\Rightarrow \frac{\mu^2}{2+\mu^2} = \frac{1}{5}$. Verify that $\mu = \frac{1}{\sqrt{2}} = 0.707$.

Verify that $\beta = \frac{\mu A_c}{2} = \frac{\mu \alpha}{2} = 4 \sqrt{2} \approx 5.657$.

Modulation index $\mu(\alpha, \beta) = \frac{2\beta}{\alpha}$.

(iv) Compute the average power of the AM signal $s(t)$ in (iii). Represent the value in the dB scale. [2]

Ans. AM signal power $= P_c \left(1 + \frac{\mu^2}{2}\right) = 128 \times \frac{5}{4} = 160 \text{ W} \approx 22 \text{ dB}$.

(v) The figure of merit of an AM receiver depends on the modulation index μ . Compute the figure of merit. Assume that the noise at receiver is small with respect to the signal. [1]

Ans. AM receiver F.O.M $= \frac{\mu^2}{\mu^2 + 2} = \frac{1}{5} = 0.2$.

Q. 2. The modulating signal is a single-tone given by $m(t) = \cos(2\pi f_m t)$ and is applied to a phase modulator with phase sensitivity k_p . The phase modulated signal is of the form $A_c \cos(2\pi f_c t + \psi(t))$.

(i) Write the general expression for the single-tone phase modulated signal $s(t)$. What is the maximum phase deviation? [3]

Ans. The single-tone phase modulated signal is given by

$$s(t) = A_c \cos(2\pi f_c t + k_p m(t)) = A_c \cos(2\pi f_c t + k_p \cos(2\pi f_m t)).$$

Maximum phase deviation $= \max\{k_p \cos(2\pi f_m t)\} = k_p$.

(ii) Let $A_c = 90 \text{ V}$ and the maximum phase deviation is 0.1 rad . For θ very small ($\ll 1 \text{ rad}$), using the approximation $\cos(\theta) \approx 1$ and $\sin(\theta) \approx \theta$, determine the simplified expression for the phase modulated signal $s(t)$. Express $s(t)$ as the sum of the carrier term, LSB term and, USB term. [2.5]

Ans. The phase modulated signal is given by $s(t) = 90 \cos(2\pi f_c t + 0.1 \cos(2\pi f_m t))$.

Using the approximations and further simplification yields

$$s(t) = 90 \cos(2\pi f_c t) - 4.5 \sin(2\pi(f_c - f_m)t) - 4.5 \sin(2\pi(f_c + f_m)t).$$

(iii) Using the three-term simplified expression for the phase modulated signal $s(t)$ obtained in (ii), determine the spectral representation $S(f)$. List all the spectral components present in $S(f)$. [5]

Ans. The continuous-time Fourier transform (CTFT) of $\cos(2\pi f_c t)$:

$$\frac{1}{2}[\delta(f - f_c) + \delta(f + f_c)]$$

The CTFT of $\sin(2\pi(f_c - f_m)t)$:

$$\frac{1}{2j} [\delta(f - f_c + f_m) - \delta(f + f_c - f_m)]$$

The CTFT of $\sin(2\pi(f_c + f_m)t)$:

$$\frac{1}{2j} [\delta(f - f_c - f_m) - \delta(f + f_c + f_m)]$$

$\therefore$ The spectral representation $S(f)$ is given by

$$\begin{aligned} S(f) = 45\{\delta(f - f_c) + \delta(f + f_c)\} - \frac{2.25}{j} [\delta(f - f_c + f_m) - \delta(f + f_c - f_m)] \\ - \frac{2.25}{j} [\delta(f - f_c - f_m) - \delta(f + f_c + f_m)]. \end{aligned}$$

Since $\frac{1}{j} = -j$, we get

$$\begin{aligned} S(f) = 45\{\delta(f - f_c) + \delta(f + f_c)\} + 2.25 j [\delta(f - f_c + f_m) - \delta(f + f_c - f_m)] \\ + 2.25 j [\delta(f - f_c - f_m) - \delta(f + f_c + f_m)]. \end{aligned}$$

Spectral components: $\pm f_c$, $\pm f_c \pm f_m$.

(iv) Determine the complex envelope of $s(t)$. Consider the general expression obtained in (i). [2]

Ans. The phase modulated signal is given by $s(t)$ is given by

$$s(t) = \Re\{A_c e^{j(2\pi f_c t + k_p m(t))}\},$$

where $\Re\{\}$ denotes the real part.

$$s(t) = \Re\{A_c e^{j(2\pi f_c t + k_p m(t))}\} = \Re\{A_c e^{j k_p m(t)} e^{j 2\pi f_c t}\}.$$

$\therefore$ The complex envelope is given by

$$\tilde{s}(t) = A_c e^{j k_p m(t)} = A_c e^{j k_p \cos(2\pi f_m t)} = A_c \cos(k_p \cos(2\pi f_m t)) + j A_c \sin(k_p \cos(2\pi f_m t)).$$

Q. 3. Consider an ideal band-pass filter (BPF) with mid-band frequency f_c . The amplitude response of the BPF is band-limited to $f_c - W \leq |f| \leq f_c + W$, $f_c \gg W$. Let $\tilde{H}(f)$ denote the low-pass equivalent filter, given by

$$\tilde{H}(f) = \begin{cases} \exp(-j 3.6 \pi f), & -W < f < W, \\ 0, & \text{elsewhere.} \end{cases}$$

(i) Suppose $W = 900 \text{ Hz}$. Determine the impulse response $\tilde{h}(t)$ of the filter. Simplify. Is the impulse response real or complex? [2.5]

Ans. Consider the following amplitude response:

$$\tilde{H}(f) = \begin{cases} \exp(-j 2 \pi f t_0), & -W < f < W, \\ 0, & \text{elsewhere.} \end{cases}$$

The impulse response $\tilde{h}(t)$ of the filter is given by

$$\tilde{h}(t) = 2W \text{sinc}(2W(t - t_0)).$$

Since $W = 900 \text{ Hz}$, $t_0 = 1.8$, we get

$$\tilde{h}(t) = 1800 \text{sinc}(1800(t - 1.8)).$$

Clearly, the impulse response $\tilde{h}(t)$ is real.

(ii) Consider the pulsed RF signal $x(t) = \{\text{rect}(t)\} \times \cos(2 \times 10^8 \pi t)$. Determine $\tilde{x}(t)$, which is the complex envelope of $x(t)$. Determine $\tilde{X}(f)$, the Fourier transform of the complex envelope. [3]

Ans. Recall that complex envelope $\tilde{x}(t) = \Re\{x_+(t) e^{-j 2\pi f_c t}\}$, where $x_+(t)$ denotes the pre-envelope.

Verify that $x_+(t) = x(t) + j \hat{x}(t) = \text{rect}(t) e^{j 2\pi f_c t}$.

$\therefore \tilde{x}(t) = \text{rect}(t)$. Verify that $\tilde{X}(f) = \text{sinc}(f)$.

(iii) The signal $\tilde{x}(t)$ is applied to the ideal filter described by $\tilde{H}(f)$. The output $\tilde{y}(t)$ is given by the following expression:

$$\tilde{y}(t) = \lambda [Si\{\mu(t - \tau_\ell)\} - Si\{\mu(t - \tau_u)\}], \text{ where } Si(v) = \int_0^v \frac{\sin y}{y} dy. \text{ Determine } \lambda, \mu, \tau_\ell \text{ and } \tau_u.$$

Ans. The output $\tilde{y}(t) = \tilde{x}(t) * \tilde{h}(t)$.

Since $\tilde{h}(t) = 2W \text{sinc}(2W(t - t_0))$ and $\tilde{x}(t) = \text{rect}(t)$,

$$\text{we get } \tilde{y}(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau = \int_{-0.5}^{0.5} 2W \text{sinc}(2W(t - t_0 - \tau)) d\tau.$$

Use the definition of $\text{sinc}(t) = \frac{\sin(\pi t)}{\pi t}$ and simplify the integral using substitution method.

Verify that

$$\tilde{y}(t) = \frac{1}{\pi} [Si\{2\pi W(t - (t_0 - 0.5))\} - Si\{2\pi W(t - (t_0 + 0.5))\}], \text{ where } W = 900 \text{ Hz, } t_0 = 1.8.$$

$$\therefore \lambda = \frac{1}{\pi}, \mu = 1800\pi, \tau_\ell = 1.3, \tau_u = 2.3.$$

(iv) Write down the relationship between $y(t)$ and $\tilde{y}(t)$. Obtain the expression for $y(t)$ using $\tilde{y}(t)$ (obtained after substituting the values of λ, μ, τ_ℓ and τ_u) from the previous part (iii). [3]

Ans. Recall that $y(t)$ is the real part of the pre-envelope $y_+(t) = y(t) + j\hat{y}(t)$, where $\hat{y}(t)$ is the Hilbert transform of $y(t)$. The pre-envelope and the complex envelope are related by

$$\tilde{y}(t) = y_+(t)e^{-j2\pi f_c t}.$$

$$\therefore y(t) = \Re\{y_+(t)\} = \Re\{\tilde{y}(t)e^{j2\pi f_c t}\}.$$

Verify that

$$y(t) = \frac{1}{\pi} [Si\{1800\pi(t - 1.3)\} - Si\{1800\pi(t - 2.3)\}] \cos(2\pi \times 10^8 t).$$

Q. 4. [Part-(A)] Consider a wide sense stationary (WSS) random process $Y(t)$. Its autocorrelation function is denoted by $R_Y(\tau)$ and power spectral density (PSD) by $S_Y(f)$.

Prove: $R_Y(\tau) = k \int_0^\infty S_Y(f) \cos(2\pi f\tau) df$, where k is a real positive constant.

Determine the value of k . [2.5]

Ans. Recall that the ACF and PSD form Fourier transform pair: $R_Y(\tau) \Leftrightarrow S_Y(f)$.

$$\therefore R_Y(\tau) = \int_{-\infty}^{\infty} S_Y(f) e^{j2\pi f\tau} df = \int_{-\infty}^{\infty} S_Y(f) \cos(2\pi f\tau) df + \int_{-\infty}^{\infty} S_Y(f) \sin(2\pi f\tau) df$$

Since PSD is even function, the second integral value is zero.

$$\therefore R_Y(\tau) = 2 \int_0^\infty S_Y(f) \cos(2\pi f\tau) df.$$

[Part-(B)] The output of an oscillator is modeled as a WSS random process, which is given by $Y(t) = \cos(2\pi F t + \Psi)$, where F and Ψ are independent continuous random variables.

Let $p_F(f)$ denote the probability density function of the random variable F . The random variable $\Psi \sim U[0, 2\pi)$.

(i) Obtain a single integral (simplified) expression for the autocorrelation function of $Y(t)$ in terms of $p_F(f)$. [4]

Ans. The ACF of $Y(t)$ is given by

$$R_Y(\tau) = E\{Y(t + \tau) Y(t)\} = \frac{1}{2} E\{\cos(4\pi Ft + 2\pi F\tau + 2\Psi)\} + \frac{1}{2} E\{\cos(2\pi F\tau)\}.$$

Since Ψ is uniformly distributed in $[0, 2\pi)$, averaging over Ψ and simplifying further, we get the following single-integral expression:

$$R_Y(\tau) = \frac{1}{2} E\{\cos(2\pi F\tau)\} = \frac{1}{2} \int_0^\infty \cos(2\pi f \tau) p_F(f) df.$$

(ii) The PSD can be expressed in the following form: $S_Y(f) = a p_F^b(f)$, where a and b are real constants. Determine a and b . Note: Use the result proved in Part-(A). [3]

Ans. From part-(A), we have

$$R_Y(\tau) = 2 \int_0^\infty S_Y(f) \cos(2\pi f \tau) df. \text{ But, } R_Y(\tau) = \frac{1}{2} \int_0^\infty p_F(f) \cos(2\pi f \tau) df.$$

Comparing the above single integral expressions, we get

$$2 S_Y(f) = \frac{1}{2} p_F(f) \Rightarrow S_Y(f) = \frac{1}{4} p_F(f). \text{ Clearly, } a = \frac{1}{4}, b = 1.$$

(iii) Suppose $p_F(f) = \frac{1}{2} \exp(-|f|)$, $f \in R$. Determine $S_Y(f)$ for all f , $S_Y(0)$ and $R_Y(0)$. [3]

Ans. $S_Y(f) = \frac{1}{4} p_F(f) = \frac{1}{8} \exp(-|f|)$. That is,

$$S_Y(f) = \begin{cases} \frac{1}{8} \exp(f), & f < 0, \\ \frac{1}{8} \exp(-f), & f \geq 0. \end{cases}$$

$$\therefore S_Y(0) = \frac{1}{8} = 0.125.$$

Verify that $R_Y(0) = 2 \int_0^\infty S_Y(f) df = \frac{1}{4} = 0.25$.

□□ **END** □□